



Part—I

SystemTap Tutorial

This is the first of a two-part series on SystemTap, a dynamic method to monitor and trace the operations of a running Linux kernel. SystemTap is useful to systems administrators, kernel developers, support engineers, researchers and students.

Who is doing the maximum read/write operations on my server?" "Can I add some debug statements in the kernel without rebuilding it and rebooting the system?"

These are questions you might have asked yourself, if you are a systems administrator or a kernel developer. Let's see what options are

available in answer to these questions:

- *Tracing*: Provides information while running, and gives a quick overview of code flow, but gives a lot of information. Tools like *strace*, *ltrace* and *ftrace* are used for tracing.
- *Profiling*: Does sampling while running, and we can do the analysis after the event. *Oprofile* is used for sampling.